ESC European Society of Cardiology

European Heart Journal (2021) **42**, 938–948
doi:10.1093/eurheartj/ehaa995

BASIC SCIENCE
Ischemic heart disease

Alarmin-activated B cells accelerate murine atherosclerosis after myocardial infarction via plasma cell-immunoglobulin-dependent mechanisms

Tin Kyaw [1,2]*, Paula Loveland [1], Peter Kanellakis[1], Anh Cao[1,2], Axel Kallies[3], Alex L. Huang [4,5], Karlheinz Peter [4,5,6], Ban-Hock Toh[2], and Alex Bobik[1,2,6]

[1]Vascular Biology and Atherosclerosis, Baker Heart and Diabetes Institute, 75 Commercial Road, Melbourne, VIC 3004, Australia; [2]Centre for Inflammatory Diseases, Department of Medicine, Medical Centre, 246 Clayton Road, Clayton, VIC 3168, Australia [3]Department of Microbiology and Immunology, University of Melbourne, 792 Elizabeth Street, Melbourne, Vic 3000, Australia; [4]Atherothrombosis and Vascular Biology, Baker Heart and Diabetes Institute, 75 Commercial Road, Melbourne, VIC 3004, Australia and [5]Department of Cardiology, Alfred Hospital, 55 Commercial Rd, Melbourne, VIC 3004, Australia[6]Department of Immunology, Central Clinical School, 99 Commercial Rd, Melbourne, VIC 3004, Australia

Received 11 February 2020; revised 30 July 2020; editorial decision 19 November 2020; accepted 29 November 2020; online publish-ahead-of-print 18 December 2020

See page 948 for the editorial comment on this article (doi: 10.1093/eurheartj/ehaa1059)

Aims

Myocardial infarction (MI) accelerates atherosclerosis and greatly increases the risk of recurrent cardiovascular events for many years, in particular, strokes and MIs. Because B cell-derived autoantibodies produced in response to MI also persist for years, we investigated the role of B cells in adaptive immune responses to MI.

Methods and results

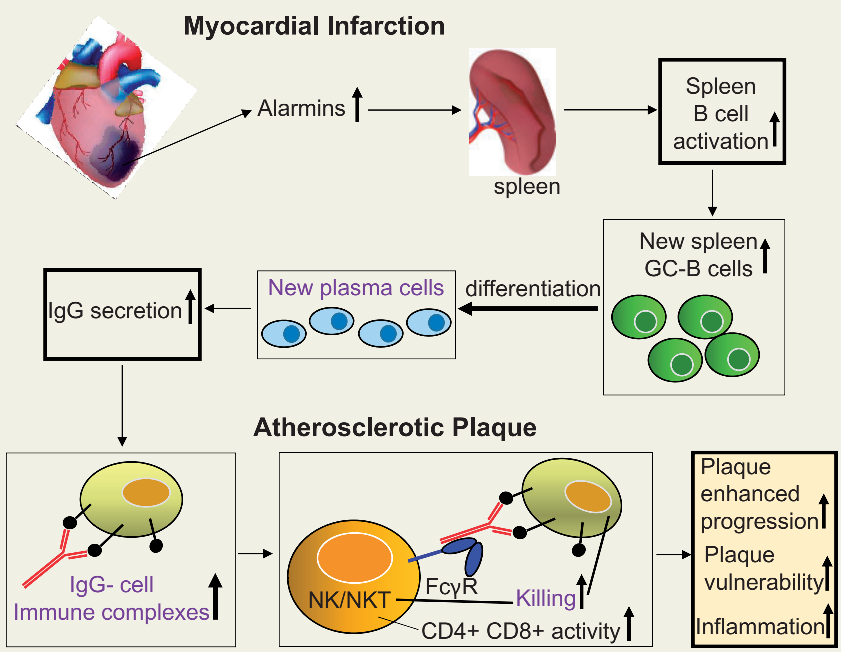
We used an apolipoprotein-E-deficient ($ApoE^{-/-}$) mouse model of MI-accelerated atherosclerosis to assess the importance of B cells. One week after inducing MI in atherosclerotic mice, we depleted B cells using an anti-CD20 antibody. This treatment prevented subsequent immunoglobulin G accumulation in plaques and MI-induced accelerated atherosclerosis. In gain of function experiments, we purified spleen B cells from mice 1 week after inducing MI and transferred these cells into atherosclerotic $ApoE^{-/-}$ mice, which greatly increased immunoglobulin G (IgG) accumulation in plaque and accelerated atherosclerosis. These B cells expressed many cytokines that promote humoural immunity and in addition, they formed germinal centres within the spleen where they differentiated into antibody-producing plasma cells. Specifically deleting Blimp-1 in B cells, the transcriptional regulator that drives their terminal differentiation into antibody-producing plasma cells prevented MI-accelerated atherosclerosis. Alarmins released from infarcted hearts were responsible for activating B cells via toll-like receptors and deleting MyD88, the canonical adaptor protein for inflammatory signalling downstream of toll-like receptors, prevented B-cell activation and MI-accelerated atherosclerosis.

Conclusion

Our data implicate early B-cell activation and autoantibodies as a central cause for accelerated atherosclerosis post-MI and identifies novel therapeutic strategies towards preventing recurrent cardiovascular events such as MI and stroke.

* Corresponding author. Tel: +61-3-85321699, Fax: +61-3-85321100, Email: tinsoe.kyaw@baker.edu.au

Graphical Abstract



Keywords Myocardial infarction • Accelerated atherosclerosis • B cells • Plasma cells • Autoantibodies

Translational perceptive

Recurrent myocardial infarctions (MIs) thought largely due to accelerated atherosclerosis account for ~40% of MIs but specific therapies are unavailable. We demonstrate that alarmins released by dying cardiomyocytes during MI activate B cells, resulting in large increases in IgG antibody-producing plasma cells. These immunoglobulins accumulate in established plaques, associate with smooth muscle and macrophages, and accelerate atherosclerosis. Preventing B-cell differentiation to plasma cells or inhibiting B-cell activation prevents MI-accelerated atherosclerosis. Therapeutically targeting B-cell activation and/or differentiation to plasma cells may be useful in preventing MI-accelerated atherosclerosis and recurrent MIs.

Introduction

After a myocardial infarction (MI), the risk of recurrent stroke and myocardial re-infarction increases greatly and remains elevated for years.[1–3] Within 5 years of a first MI, 34% will experience one or more recurrent, frequently fatal MIs.[4] Accelerated progression of non-culprit atherosclerotic lesions to vulnerable plaques following MI appears to account for such recurrent events.[5,6] However, cellular/molecular mechanisms responsible for this long-term increased risk are largely unknown, although MI in humans aggravates inflammation in atherosclerotic lesions.[7]

Myocardial infarction initiates multiphase activation of the immune system, resulting in a rapid sterile inflammatory response. This in turn stimulates the release of haematopoietic stem and progenitor cells from bone marrow niches, which seed the spleen ultimately boosting monocyte production.[6] During MI, alarmins, such as heat shock protein-60 (hsp60), high mobility group box-1 (HMGB-1), and mitochondrial DNA (mtDNA) are released, activating both T and B cells.[8,9] Activated B cells contribute to early myocardial injury following infarction by triggering monocyte mobilization[10] and later, increase antibody production in response to antigens released during MI. Antibodies detected following MI in humans include anti-endothelial[11] and anti-transglutaminase antibodies,[12] both capable of exerting pro-atherogenic effects. Anti-endothelial antibodies increase monocyte adhesion to endothelial cells, up-regulate the adhesion molecules E-selectin, intercellular-adhesion-molecule-1 and vascular-cell-adhesion-molecule-1, and increase secretion of interleukin-6 (IL-6)[13] associated with development of unstable angina.[14] Anti-transglutaminase antibodies are functional,[15] and inhibit transglutaminase activity[16] and apoptotic cell clearance by macrophages.[17] In atherosclerotic lesions, this reduction in efferocytosis increases plaque inflammation and necrotic core development, effects that lead to plaque rupture and acute thrombotic cardiovascular events.[18,19] As activated B cells can differentiate into long-lived antibody-producing plasma cells (PCs), pathogenic antibodies generated in response to MI will persist for years and continuously drive atherosclerosis via inflammatory mechanisms, greatly increasing risk of life-threatening recurrent cardiovascular events.

We hypothesized that long-term recurrent cardiovascular events seen after an initial MI are due to antibodies produced by PCs that have differentiated from B cells activated by alarmins and antigens. Here we confirm exacerbation of atherosclerosis after MI and specifically identify key roles for alarmins in activating B cells during MI; we also demonstrate that activated B cells differentiate to antibody-producing PCs, which exacerbate progression of established atherosclerosis to plaques with characteristics of plaque instability.

Methods

A detailed description of the methods is provided in the Supplementary material online.

Results

B cells contribute to myocardial infarction-accelerated atherosclerosis

To determine the role of B cells in MI-accelerated atherosclerosis, ApoE$^{-/-}$ mice fed a high-fat diet (HFD) for 6 weeks were subjected to left coronary artery ligation to induce MI or sham operation and then anti-CD20 antibody treatment. Anti-CD20 antibody depleted B cells by >90% (Supplementary material online, *Figure S1A* and *B*). Nine weeks after MI, IgG within atherosclerotic lesions of infarcted mice was increased 3.5-fold (*Figure 1A*) without affecting plasma lipids and bodyweight (Supplementary material online, *Figure S1C* and *D*). B-cell depletion normalized lesion IgGs to those in sham group (*Figure 1A*). Only small changes in plasma IgG level (~20%) were observed with B-cell depletion (Supplementary material online, *Figure S2A*), consistent with production of specific antibodies in response to MI which target established atherosclerotic lesions. To confirm that antibodies were produced in response to the MI, we next examined antibody-producing PCs in spleens. Spleen PCs were increased nearly six-fold by MI and this increase was prevented by anti-CD20 antibody treatment (*Figure 1B*). As IgG-immune complexes activate Fcγ receptors on immune cells, e.g. CD16 expressed by natural killer T (NKT) cells,[20] promoting atherosclerotic lesions via cytotoxic mechanisms,[21] we investigated whether post-MI atherosclerotic lesions contained greater numbers of apoptotic cells. Lesion apoptotic cell numbers were nearly 2.5-fold greater following MI, compared with sham-operated group (*Figure 1C*); depleting B cells in MI mice attenuated this increase in apoptotic cells. As lesion apoptotic cells contribute to necrotic core development, we examined the effects of MI and B-cell depletion on lesion necrotic core size. The latter were doubled in MI mice, compared with sham-operated mice and this increase was prevented by B-cell depletion (*Figure 1D*). Lesion macrophage densities were largely unaffected by MI and B-cell depletion (*Figure 1E*). Myocardial infraction increased atherosclerotic lesion size by nearly 40% and this was prevented by B-cell depletion (*Figure 1F*). Plasma total cholesterol, VLDL/LDL, HDL, and bodyweights were unaffected (Supplementary material online, *Figure S1C* and *D*). This indicates that after MI, B cells contribute to development of larger and more advanced atherosclerotic lesions with characteristics of vulnerability. The observed effects of B-cell depletion were not due to any indirect effect on MI size as infarct size was unaffected by anti-CD20 treatment (*Figure 1G*). We specifically avoided affecting MI size by commencing treatment later, 1 week rather than 1 h after MI, which affects infarct size by modulating early monocyte responses.[10]

Myocardial infarction-B cells promote the development of inflammatory, vulnerable plaques

To confirm that MI-activated B cells contribute to accelerated atherosclerosis, we adoptively transferred spleen B cells (MI-B cells) isolated from mice 1 week after MI into atherosclerotic mice previously fed an HFD for 10 weeks but not subjected to MI. One week after MI was chosen as the time for isolation as B-cell activation was increased at this time and reverted back to pre-MI levels by 14 days (Supplementary material online, *Figure S2B*). Six weeks later, lesion IgGs in mice receiving MI-B cells were doubled compared with those that received B cells from sham-operated mice (*Figure 2A*). IgGs produced by MI-B cells exhibited greater binding avidity for endothelial cells, vascular smooth muscle cells, and macrophages in lesions (Supplementary material online, *Figure S3*). Lesion TUNEL-positive apoptotic cell numbers were also increased (*Figure 2B*) as was necrotic core size (*Figure 2C*), consistent with effects of B-cell deletion after MI (*Figure 1C and D*). Macrophages respond to IgG-immune complexes by secreting tumour necrosis factor alpha (TNF-α) following activation of Fcɣ receptors.[22,23] Thus, we determined if lesion TNF-α levels were also increased following MI-B cell transfer. Lesion TNF-α levels were greatly increased, nearly six-fold (*Figure 2D*), in agreement with TNF-α levels increased in MI mice and reduced when B cells were depleted (*Figure 2E*). Lesion CD4+ and CD8+ T cells were also increased following MI-B cell transfer, similar to increases after MI (*Figure 2F and G*), consistent with findings that monocyte/macrophage activation via Fcɣ receptors induces expression of chemokines macrophage-inflammatory protein (MIP)-1α and MIP-1β[24]; MIP-1β stimulates CD4+ T-cell chemotaxis whilst MIP-1α stimulates CD8+ T-cell chemotaxis.[25] Myocardial infarction-B cell transfer also increased lesion size by nearly 40% compared with B cells from sham mice or saline injection (*Figure 2H*). These findings unequivocally demonstrate that MI-activated B cells accelerate atherosclerosis, increasing lesion size, inflammation, and accumulation of immunoglobulins promoting plaque vulnerability/instability.

Myocardial infarction stimulates development of spleen germinal centres and antibody-producing plasma cells

Given the close association between IgG antibodies, increased plaque vulnerability and size after MI and their relation to B cells, we next examined how spleen B cells were affected by MI. Seven days after MI, splenic B cells greatly elevated their production of cytokines that promote humoural immunity, including TNF-α for sustained antibody production,[26] IL-6 for germinal centre (GC) initiation,[24] IL-12 for dendritic cell-induced naïve B-cell differentiation,[27] and IL-18 to

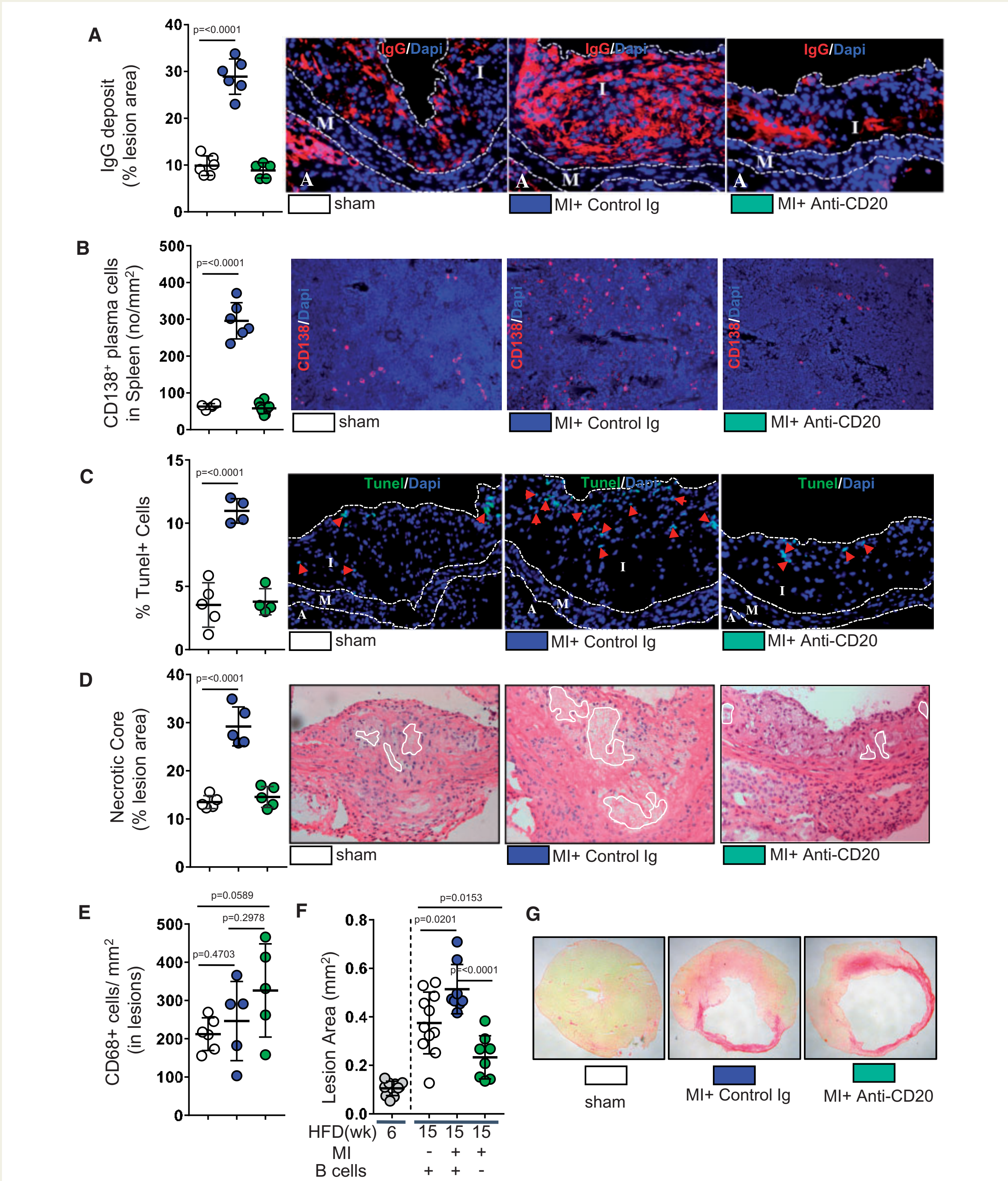


Figure 1 B-cell depletion greatly attenuates acceleration of established atherosclerosis. Hyperlipidaemic ApoE$^{-/-}$ mice fed a high-fat diet for 6 weeks were subjected to myocardial infarction and then 1 week later treated with either anti-CD20 or control antibody for 8 weeks; sham-operated non-infarcted mice were treated with saline. (*A*) Myocardial infarction-induced increase in plaque IgGs is completely prevented by anti-CD20 treatment. (*B*) Anti-CD20 treatment greatly attenuates the myocardial infarction-induced increase in antibody-producing plasma cells in spleen. Increase in (*C*) apoptotic cells and (*D*) necrotic core size following myocardial infarction is completely prevented following B-cell depletion. (*E, F*) Plaque macrophage numbers are unaffected but myocardial infarction-induced increase in plaque size is attenuated by anti-CD20 treatment that does not affect (*G*) infarct size. Data presented as mean ± SD of 2–3 independent experiments. *n* = 5–12/group. One-way ANOVA followed by Dunnett's multiple comparison test.

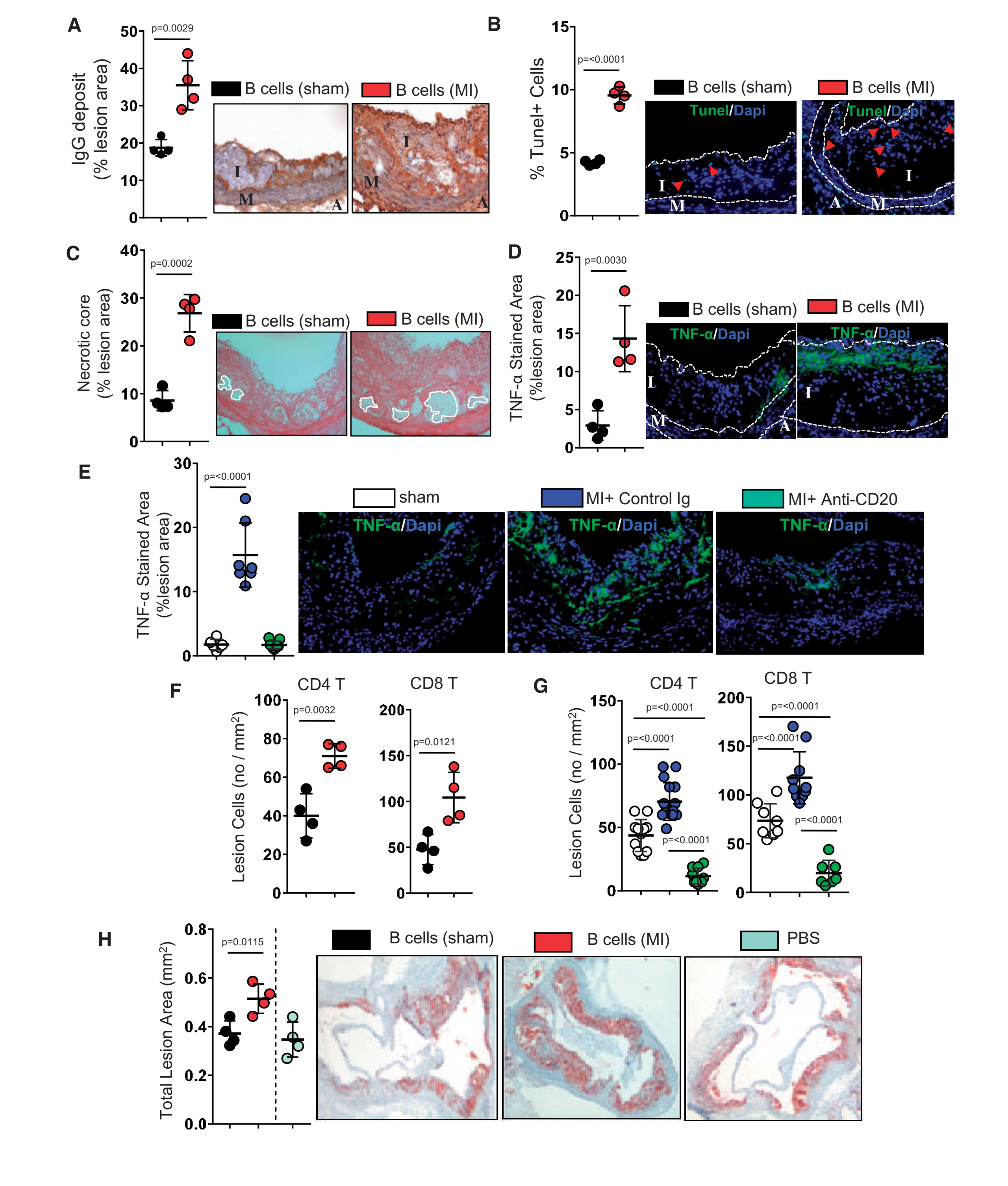


Figure 2 Adoptively transferred B cells from infarcted mice accelerate progression of established atherosclerosis. Wild-type mice were subjected to myocardial infarction. One week later splenic B cells were isolated and transferred into 10-week high-fat diet fed ApoE$^{-/-}$ mice; control mice received B cells from non-infarcted mice. Six weeks later recipient mice were killed. Transfer of B cells from infarcted mice greatly increases (*A*) lesion IgG accumulation, (*B*) plaque Tunel+ apoptotic cell numbers, and (*C*) necrotic cores. TNF-α expression, CD4+ and CD8+ T cells were increased in (*D, F*) myocardial infarction-B cell recipient mice and (*E, G*) anti-CD20 antibody-treated mice. (*H*) Myocardial infarction-B cells increase plaque size compared with non-myocardial infarction-B cells or saline injections. Data presented as mean ± SD of 2–3 independent experiments. *n* = 5–12/group. Two-tailed unpaired Student's *t*-test or One-way ANOVA followed by Dunnett's multiple comparison test.

augment GC formation[28] with their mRNA expression increased by four-fold, approximately seven-fold, approximately seven-fold and four-fold, respectively (*Figure 3A*). B-cell expression of CCR7, which enables antigen-activated B cells to migrate to the boundary between B- and T-cell zones and interact with T cells before competing for entry into GCs,[29,30] was also elevated approximately six-fold (*Figure 3A* and *B*). Next, we assessed whether GCs where B cells differentiate into PCs were increased. Clusters of GC B cells identified by PNA staining were greater in spleens of infarcted mice (*Figure 3C*). Seven days after MI, spleen PCs were also increased by ~8.5-fold (*Figure 3D*).

To confirm that antibody-producing PCs were responsible for accelerated atherosclerosis, we examined the effects of preventing B-cell differentiation to PCs on accelerated atherosclerosis. Initiation of PC differentiation within GCs involves induction of Blimp-1, the PC transcriptional regulator.[31] We induced MI in Blimp-1$^{+/+}$CD23-CRE mice and Blimp-1$^{fl/fl}$CD23-CRE mice, whose B cells are deficient in Blimp-1.[32] Mature B cells, immature B T1 and T2 cells, marginal zone B cells and B1a cells are normal in these mice.[32] Seven days after MI, spleen B cells were isolated and transferred into 10 week HFD-fed atherosclerotic ApoE$^{-/-}$ mice and atherosclerotic lesions compared 6 weeks later. In mice given Blimp-1-deficient MI-B cells, lesions were ~50% smaller than in mice whose received wild-type MI-B cells (*Figure 3E*), confirming the role of antibody-producing PCs in MI-accelerated atherosclerosis.

To determine whether plaque IgGs exert effects through antibody (IgG)-dependent cellular cytotoxicity via FcγRIII (CD16), we performed co-localization stainings on plaque NK, CD8+ T cells and macrophages. CD16 was detected on all three immune cell types (*Figure 3F*) supporting our hypothesis that this mechanism likely contributes to increased plaque cell death, reflected by increases in necrotic core size and inflammation after MI.

B cells activated by alarmins released during myocardial infarction accelerate atherosclerosis

To determine whether alarmin release is associated with B-cell activation, we examined peripheral CD86+ B cells in humans with MI within the first 72 h of their admission to emergency, and compared levels with a control group consisting of patients who underwent elective coronary angiography. Flow cytometry analysis indicated a five-fold increase in CD86+ B cells in MI patients (*Figure 4A*). Similarly, CD86+ and CCR7+ B cells were increased in mice 7 days after MI, by ~40% and 100%, respectively (*Figure 4B*). To examine relationships between B cells and alarmins in mice, we assessed HMGB-1, hsp60, and mtDNA as well as spleen B-cell activation. Plasma HMGB-1 increased rapidly, peaking 7 days after infarction and returning to normal by 21 days; plasma mtDNA levels followed a similar time course whilst plasma hsp60 elevation was more rapid and sustained for 5 days before returning to control levels by 21 days (*Figure 4C–E*). We also determined plasma HMGB-1, hsp60, and mtDNA in patients with MI. As expected, plasma alarmin levels were increased in patients with MI (*Figure 4F*) and high levels of plasma HMGB-1, hsp60, and mtDNA are independently associated with mortality in MI patients.[33–35] Yet, we did not find any increase in anti-HMGB-1 and anti-hsp60 antibodies in humans (*Figure 4G*) or mice (Supplementary material online, *Figure S2C*) within a week after MI and it is likely because IgG-producing PCs only commence to appear 7–10 days after antigen exposure.[36] Despite small changes in plasma IgGs, B-cell depletion did not affect plasma anti-HMGB-1 and anti-hsp60 antibodies in MI mice at completion of 15-week HFD (Supplementary material online, *Figure S2D*). Spleen B cells were highly activated 7 days after MI, with CD86+ B cells increased by four-fold (*Figure 4H*). As these alarmins, notably HMGB-1 and hsp60 exert their effects by interacting with toll-like receptors (TLRs), specifically TLR4[37,38] and mtDNA interacts with TLR9,[39] we assessed the significance of their activation for accelerated atherosclerosis. Wild-type mice and mice deficient in MyD88, a canonical adaptor protein for inflammatory signalling pathways downstream of TLR4 and TLR9,[40] were subjected to MI. Then spleen B cells isolated 7 days later were transferred into 10-week HFD-fed atherosclerotic ApoE$^{-/-}$ mice; 6 weeks after cell transfers, we assessed effects on lesion size. Lesion size was nearly 60% larger (*Figure 4I*) in atherosclerotic mice that received MI-B cells from wild-type mice compared with mice that received MI-B cells from MyD88-deficient mice. These findings demonstrate the importance of alarmin-activated B cells to MI-accelerated atherosclerosis.

Discussion

Our findings indicate that B-cell-driven immune mechanisms are fundamental to adaptive immune responses in MI-accelerated atherosclerosis. Following activation by MI-derived alarmins, B cells differentiate into antibody-producing PCs within the spleen and remotely drive atherosclerosis, by producing IgG antibodies that accumulate in pre-existing atherosclerotic plaques, increasing local inflammation, and accelerating plaque progression towards a vulnerable phenotype. Plasma cells are generally long-lived, produce antibodies for many years,[11,41] and thereby most likely account for much of the long-term increased risk of recurrent MIs and strokes after a first MI.

We used 'loss and gain of function' approaches to assess the role of MI-activated B cells in accelerating established atherosclerosis. Loss of function involved depleting B cells using an anti-CD20 antibody. This strategy prevented MI-stimulated increases in spleen antibody-producing PCs and lesion increases in IgG, apoptotic cells, necrotic core size, and effects on plaque size. The somewhat larger effect of B-cell depletion on lesion size after MI compared with non-MI controls is not unexpected given that B cells also contribute to *normal* progression of atherosclerosis.[42] Transfer of purified MI-activated B cells into non-MI atherosclerotic mice confirmed their role in MI-accelerated atherosclerosis. Transfer of these cells also increased lesion IgGs, apoptotic cells, necrotic core size, and lesion size. These two approaches provide unequivocal evidence for MI-activated B cells accelerating progression of established lesions, towards a vulnerable phenotype.

Autoantibodies have been strongly associated with atherogenesis,[43,44] in particular in those with autoimmune disorders.[44] Our finding that MI greatly enhances accumulation of IgG indicates an important role for autoantibodies and B-cell-derived PCs generated in response to MI. Blimp-1 is a transcriptional regulator that drives terminal differentiation of B cells into antibody-producing PCs.[31]

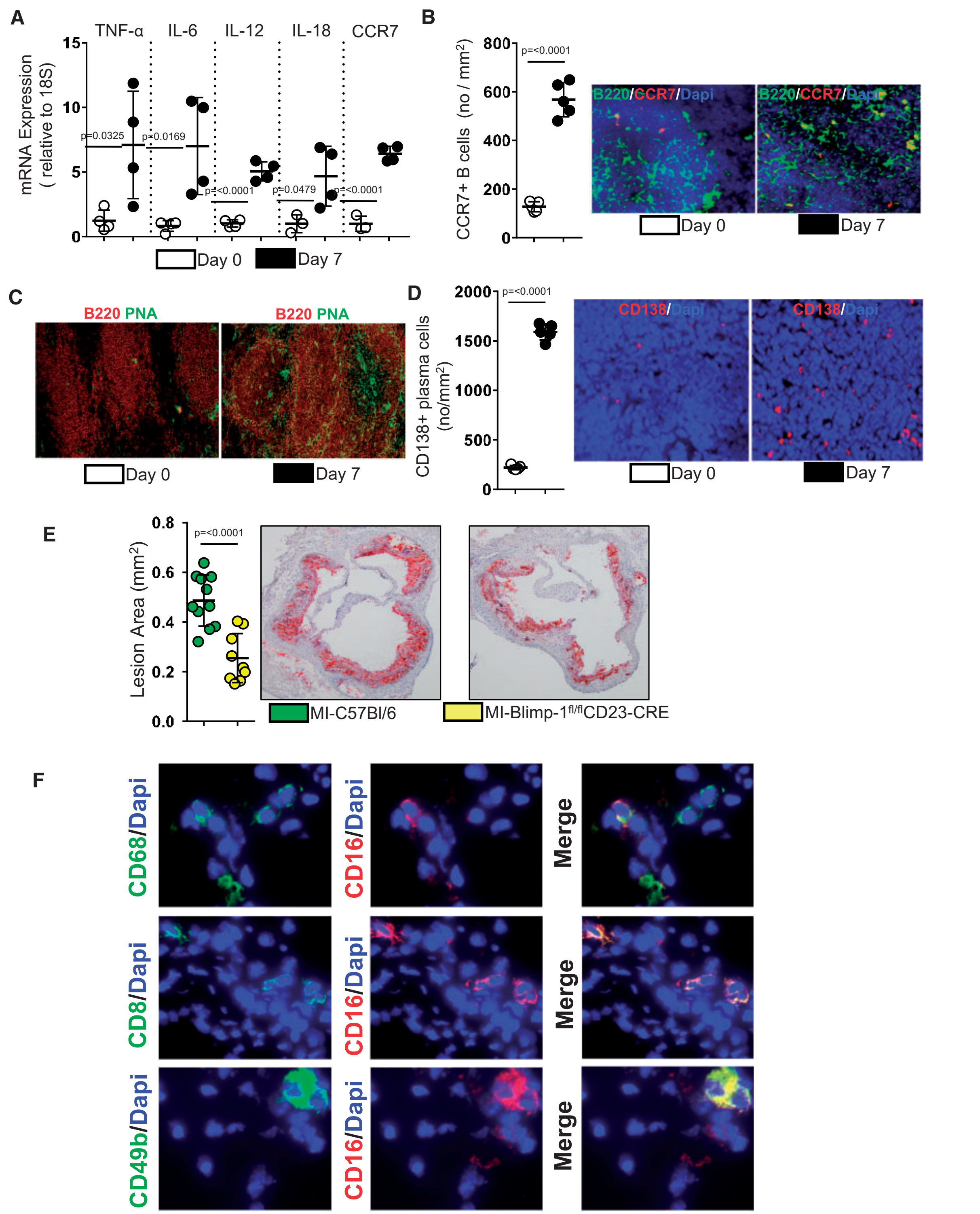


Figure 3 Myocardial infarction activates B cells to differentiate into antibody-producing plasma cells and to accelerate progression of pre-existing atherosclerosis. Seven days after myocardial infarction (*A*) spleen B cells express elevated mRNAs encoding multiple cytokines: TNF-α, interleukin-6, interleukin-12 and interleukin-18 and CCR7, all important in B-cell humoural immunity and (*B*) CCR7+ B cells in spleen are also increased. (*C*) Increased numbers of PNA+ germinal centre B cells within spleen are also increased 7 days after infarction as (*D*) are spleen plasma cells. (*E*) B cell transfer from infarcted Blimp-1$^{fl/fl}$CD23-CRE mice attenuates accelerated atherosclerosis compared with B cell transfers from infarcted wild-type mice. (*F*) Immunofluorescence staining indicates that lesion IgG deposits co-localize with Fcɣ receptor (CD16)-expressing CD68+ monocytes/macrophages, CD8+ T cells and CD49b+ NK/NKT cells. Data presented as mean ± SD of 2–3 independent experiments. *n* = 4–12/group. Two-tailed unpaired Student's *t*-test.

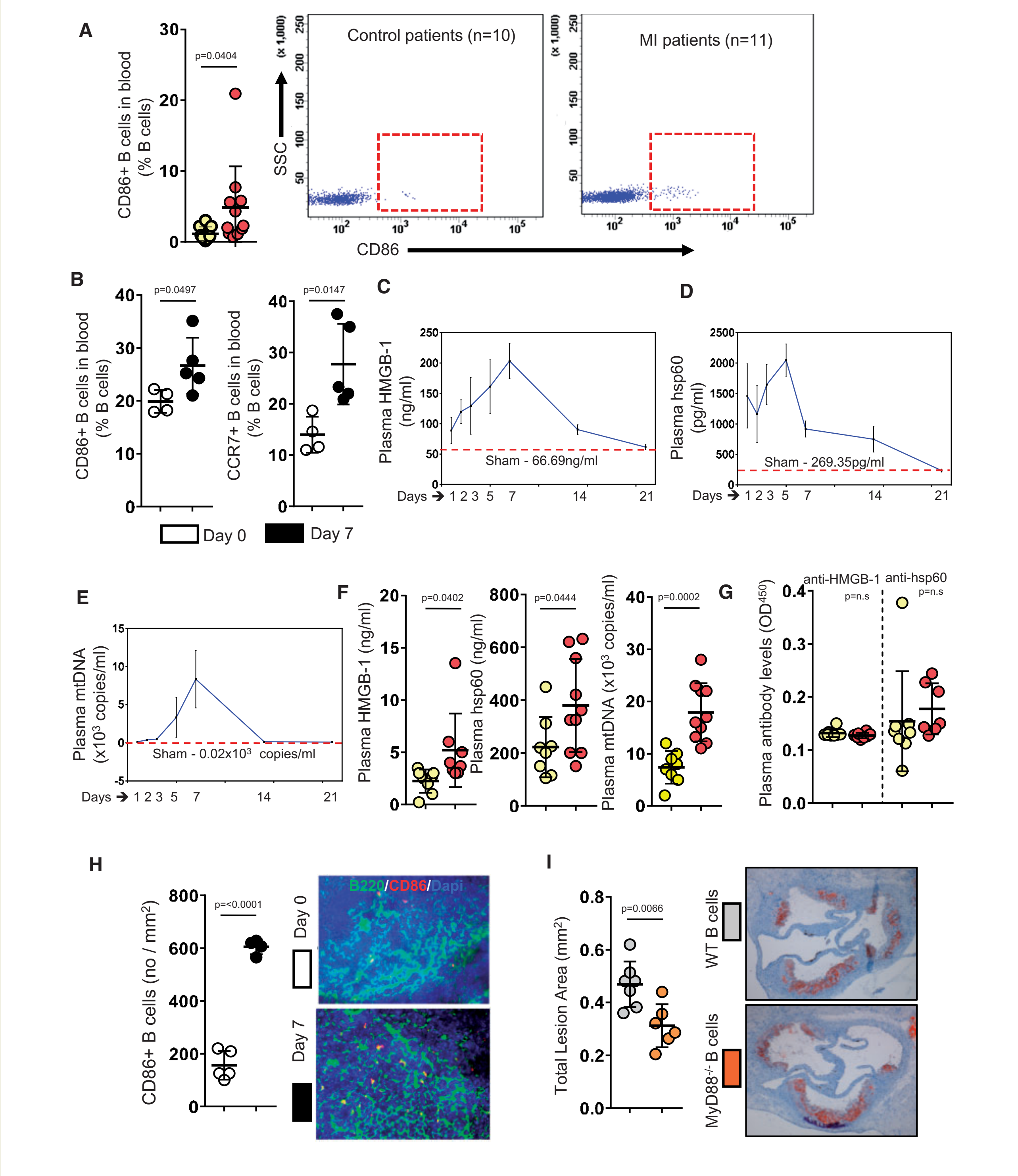


Figure 4 Alarmins released by infarcted myocardium activate B cells, which in turn accelerate atherosclerosis. (*A*) B220+ CD86+ B cells are increased in humans early after myocardial infarction. (*B*) B220+ CD86+ and B220+ CCR7+ B-cell numbers are increased in mice 7 days after myocardial infarction. (*C–E*) Time course increases in plasma high mobility group box-1, heat shock protein-60, and mitochondrial DNA after myocardial infarction in mice levels in mice following myocardial infarction. Similar to animal observations, myocardial infarction patients also had increased (*F*) plasma high mobility group box-1, heat shock protein-60, and mitochondrial DNA within 72 h after myocardial infarction, however (*G*) anti-high mobility group box-1 and anti-heat shock protein-60 antibody levels were unaffected. Myocardial infarction greatly increases (*H*) C86+ B-cell numbers in spleens of mice. (*I*) Adoptive transfer of spleen B cells from infarcted MyD88-deficient mice greatly fails to accelerate atherosclerosis compared with those from infarcted wild-type mice. Data presented as mean ± SD of 2–3 independent experiments. n = 5–12/group. Two-tailed unpaired Student's *t*-test.

Deleting B-cell-specific Blimp-1 attenuated the effects of MI on atherosclerosis, confirming the importance of antibodies in promoting atherosclerosis. Plasma cells do not accumulate in atherosclerotic lesions but rather survive in niches within bone marrow[45] or even in the adventitia within artery tertiary lymphoid structures[46] where they produce antibodies, influencing plaque progression. In addition to activating macrophages to secrete TNF-α and recruiting T cells, IgG-immune complexes interact with Fcɣ receptor III activating immune cells, including NKT and NK cells.[42] Natural killer T cells promote atherosclerosis via cytotoxic mechanisms that include increasing the number of lesion apoptotic cells and lesion necrotic core size,[47] effects identical to those observed after MI. We hypothesize that NKT/NK cells activated by IgG antibodies produced by PCs initiate a cascade of events that includes activated NKT cells enhancing CD4+ and CD8+ T-cell responses in the plaque,[48] thus accelerating atherosclerosis after MI; NKT cells, CD4+ and CD8+ T cells are present not only in mouse atherosclerotic lesions but are also highly abundant in advanced human plaques.[49–51]

As to mechanisms by which MI activates B cells, our studies indicate an important role for factors released by infarcted tissue, specifically alarmins. They interact with TLRs on B cells activating B cells to express cytokines required for promoting antibody production[24–28,52] and the chemokine receptor CCR7 required for their migration to the T–B cell border,[29,30] before forming GCs and differentiating into antibody-producing PCs. These latter effects are consistent with increased autoantibodies observed in humans after MI, specifically anti-endothelial and anti-transglutaminase antibodies,[11,12] which exert pro-atherogenic effects.[14,17,19] Inhibiting signalling through TLRs by deleting MyD88 from B cells completely prevents their ability to accelerate atherosclerosis after MI. TLRs/MyD88 internal signalling is important in antigen-presenting cells such as dendritic cells and B cells, as a deficiency in MyD88 impaired humoural responses against infections.[53–55] Together with our data that Blimp-1-deficient MI-B cells fail to augment pre-existing atherosclerosis and lesion IgGs, our data collectively provide evidence that immunoglobulins generated in response to MI, contribute to MI-accelerated atherosclerosis. Our finding that B-cell depletion only slightly reduced plasma IgG levels after MI whilst completely preventing plaque IgG increases suggests that local rather than humoural responses may be more important in accelerating atherosclerosis. Local IgG could be derived from artery tertiary lymphoid organ structures (ATLOS) associated with atherosclerotic plaques[46]; ATLOS are located within the adventitia and contain T and B cells as well as long-lived IgG-producing PCs.[46] However, we did not find any significant decrease in anti-HMGB-1 and hsp60 antibodies in anti-CD20 antibody-treated mice, despite less aggravated atherosclerosis and decreased total plasma IgG levels. Alarmins released in response to cell injuries or cell death are endogenous immune-activating proteins/peptides and have shown biological effects; many of which remain to be identified.[56] Our group and others reported that hsp60, HMBG-1, and mtDNA are detected in both human and mouse atherosclerotic lesions and promote atherosclerosis.[57–62] The myocardium contains many different cell types in addition to cardiomyocytes and fibroblasts, which are common to the heart and atherosclerotic plaques including endothelial cells, vascular smooth muscle cells, macrophages, and small populations of T and B cells.[63] Antigens derived from such dying myocardium and subsequent antibodies generated to these antigens have the potential to influence the function of these cells within atherosclerotic plaques which may accelerate atherogenesis. The important role of newly generated antibodies, particularly IgGs, in response to myocardial injuries is highlighted in MI-accelerated atherosclerosis and an increase in serum IgG has been documented in MI survivors.[64] The roles of antibodies against specific alarmins remain to be further investigated.

To demonstrate the translational relevance of our experimental data in mice, we show increased plasma HMGB-1, hsp-60, and mtDNAs in MI patients. Furthermore, MI patients exhibited activated B cells by flow cytometry. These data indicate the translational relevance of alarmins and B-cell activation in patients after MI. Whilst MI survivors are at increased risk of recurrent MIs and ischaemic strokes,[1–3] stroke victims are also prone to recurrent ischaemic strokes and MIs.[65,66] Accelerated inflammation due to tissue injury seems to be an important factor in accelerating pre-existing atherosclerotic lesions as notably non-cardiac, non-neurological surgical procedures also increase risk of acute and chronic post-operative ischaemic myocardial and cerebral events.[67,68] B-cell activation and pathogenic antibodies impair the recovery phases and remodelling process in MIs and spinal cord injuries.[69,70]

Our study identifies novel pathways that may be targeted to prevent MI-accelerated atherosclerosis. First, transient B-cell depletion following MI during the time which alarmins are elevated may prevent early B-cell activation and its downstream consequences on antibody production and atherosclerosis; anti-CD20 therapy is currently being trialled for limiting infarct size in patients with acute ST-elevation MI (ClinicalTrials.gov Identifier: NCT03072199) and could be extended to accelerated atherosclerosis subject to safety evaluations. In mice, a single injection of anti-CD20 after MI completely prevents the acceleration of atherosclerosis (Supplementary material online, *Figure S4*). Preventing TLR activation on B cells by indirectly targeting MyD88 may also be an effective strategy. IRAK4, a serine–threonine kinase is essential for TLR signalling through MyD88; inhibiting IRAK4 effectively disrupts the TLR-MyD88 signalling pathway.[71] IRAK4 inhibitors are in clinical trials for rheumatoid arthritis (ClinicalTrials.gov Identifier: NCT02996500) and skin inflammatory conditions (ClinicalTrials.gov Identifier: NCT04092452).

Our study has a number of limitations. First, experiments were performed in mice and it is not known whether all downstream signals following MI in mice are the same in humans. Despite this, the study of patients with MI, although limited in its extent, indicates that signals which activate B cells are most likely the same in humans. Also, we could only study effects on atherosclerosis and not associated recurrent MIs. Plaque rupture is mostly responsible for MIs and whilst a number of plaque rupture models have recently been developed,[72,73] there is no evidence to indicate that mechanisms of plaque rupture in mouse plaques are the same as those in humans. Unlike humans, mouse models require either genetic or surgical manipulation of atherosclerotic vessels to induce plaque rupture.[72,73] Thus, we assessed plaque phenotypes only indirectly, via histological measures that are known to be associated with plaque instability in humans. Although our current study provides an indication for translational relevance of our findings, more extensive clinical studies involving larger number of patients are required. Last but not least, our inability to detect anti-hsp60 and anti-HMGB-1 antibodies in MI patients should also be considered a limitation. Others have reported that following MI in humans plasma total IgG levels are increased,[64] similarly anti-endothelial cell[11] and anti-transglutaminase[12] antibodies are also increased, suggesting a need to search for additional antibodies generated in response to antigens released from dying myocardium following MI that are also present in atherosclerotic plaques.

In conclusion, we identify B cells as an important accelerator of established atherosclerosis following MI. Our findings indicate critical roles for alarmins, released by infarcted myocardium, in stimulating B cells, which then differentiate into PCs and produce autoantibodies that accelerate atherosclerosis. We also identify a 'therapeutic window' for preventing B-cell activation and accelerated atherosclerosis. These findings provide new mechanistic insights and point to novel therapeutic strategies towards prevention of the secondary acceleration of atherosclerosis after MI and its associated recurrent cardiovascular events.

Supplementary material

Supplementary material is available at *European Heart Journal* online.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Funding

This work was supported by grants from the National Health and Medical Research Council of Australia (Investigator grant to K.P., Project grant 1002333 to B.-H.T. and A.B.) and the Contributing to Australian Scholarship and Science (CASS) Foundation (SM/13/4956 to T.K.) and in part by the Victorian State Government Operational Infrastructure Support grant.

Conflict of interest: none declared.

References

1. Kannel WB, Sorlie P, McNamara PM. Prognosis after initial myocardial infarction: the Framingham study. *Am J Cardiol* 1979;**44**:53–59.
2. Jernberg T, Hasvold P, Henriksson M, Hjelm H, Thuresson M, Janzon M. Cardiovascular risk in post-myocardial infarction patients: nationwide real world data demonstrate the importance of a long-term perspective. *Eur Heart J* 2015;**36**:1163–1170.
3. Witt BJ, Brown RD Jr, Jacobsen SJ, Weston SA, Yawn BP, Roger VL. A community-based study of stroke incidence after myocardial infarction. *Ann Intern Med* 2005;**143**:785–792.
4. Kochar A, Chen AY, Sharma PP, Pagidipati NJ, Fonarow GC, Cowper PA, Roe MT, Peterson ED, Wang TY. Long-term mortality of older patients with acute myocardial infarction treated in US clinical practice. *J Am Heart Assoc* 2018;**7**:e007230.
5. Han Y, Jing J, Tu S, Tian F, Xue H, Chen W, Chen J, Reiber JH, Chen Y. ST elevation acute myocardial infarction accelerates non-culprit coronary lesion atherosclerosis. *Int J Cardiovasc Imaging* 2014;**30**:253–261.
6. Dutta P, Courties G, Wei Y, Leuschner F, Gorbatov R, Robbins CS, Iwamoto Y, Thompson B, Carlson AL, Heidt T, Majmudar MD, Lasitschka F, Etzrodt M, Waterman P, Waring MT, Chicoine AT, van der Laan AM, Niessen HW, Piek JJ, Rubin BB, Butany J, Stone JR, Katus HA, Murphy SA, Morrow DA, Sabatine MS, Vinegoni C, Moskowitz MA, Pittet MJ, Libby P, Lin CP, Swirski FK, Weissleder R, Nahrendorf M. Myocardial infarction accelerates atherosclerosis. *Nature* 2012;**487**:325–329.
7. Joshi NV, Toor I, Shah AS, Carruthers K, Vesey AT, Alam SR, Sills A, Hoo TY, Melville AJ, Langlands SP, Jenkins WS, Uren NG, Mills NL, Fletcher AM, van Beek EJ, Rudd JH, Fox KA, Dweck MR, Newby DE. Systemic atherosclerotic inflammation following acute myocardial infarction: myocardial infarction begets myocardial infarction. *J Am Heart Assoc* 2015;**4**:e001956.
8. Li Y, Si R, Feng Y, Chen HH, Zou L, Wang E, Zhang M, Warren HS, Sosnovik DE, Chao W. Myocardial ischemia activates an injurious innate immune signaling via cardiac heat shock protein 60 and Toll-like receptor 4. *J Biol Chem* 2011;**286**:31308–31319.
9. Tian Y, Charles EJ, Yan Z, Wu D, French BA, Kron IL, Yang Z. The myocardial infarct-exacerbating effect of cell-free DNA is mediated by the high-mobility group box 1-receptor for advanced glycation end products-Toll-like receptor 9 pathway. *J Thorac Cardiovasc Surg* 2019;**157**:2256–2269.e3.
10. Zouggari Y, Ait-Oufella H, Bonnin P, Simon T, Sage AP, Guerin C, Vilar J, Caligiuri G, Tsiantoulas D, Laurans L, Dumeau E, Kotti S, Bruneval P, Charo IF, Binder CJ, Danchin N, Tedgui A, Tedder TF, Silvestre JS, Mallat ZB. lymphocytes trigger monocyte mobilization and impair heart function after acute myocardial infarction. *Nat Med* 2013;**19**:1273–1280.
11. Elsheikh E, Sylven C, Henareh L. Anti-endothelial cell antibodies are increased in patients with previous myocardial infarction. *Scand Cardiovasc J* 2010;**44**:255–259.
12. Di Tola M, Barilla F, Trappolini M, Palumbo HF, Gaudio C, Picarelli A. Antitissue transglutaminase antibodies in acute coronary syndrome: an alert signal of myocardial tissue lesion? *J Intern Med* 2007;**263**:43–51.
13. Del Papa N, Raschi E, Moroni G, Panzeri P, Borghi MO, Ponticelli C, Tincani A, Balestrieri G, Meroni PL. Anti-endothelial cell IgG fractions from systemic lupus erythematosus patients bind to human endothelial cells and induce a pro-adhesive and a pro-inflammatory phenotype in vitro. *Lupus* 1999;**8**:423–429.
14. Farsi A, Domeneghetti MP, Brunelli T, Gori AM, Fedi S, Gensini GF, Giglioli C, Prisco D, Passaleva A, Meroni PL, Del Papa N, Abbate R. Activation of the immune system and coronary artery disease: the role of anti-endothelial cell antibodies. *Atherosclerosis* 2001;**154**:429–436.
15. Caputo I, Barone MV, Lepretti M, Martucciello S, Nista I, Troncone R, Auricchio S, Sblattero D, Esposito C. Celiac anti-tissue transglutaminase antibodies interfere with the uptake of alpha gliadin peptide 31-43 but not of peptide 57-68 by epithelial cells. *Biochim Biophys Acta* 2010;**1802**:717–727.
16. Esposito C, Paparo F, Caputo I, Rossi M, Maglio M, Sblattero D, Not T, Porta R, Auricchio S, Marzari R, Troncone R. Anti-tissue transglutaminase antibodies from coeliac patients inhibit transglutaminase activity both in vitro and in situ. *Gut* 2002;**51**:177–181.
17. Sóñora C, Mourglia-Ettlin G, Calo G, Hauk V, Ramhorst R, Hernández A, Leirós CP. Anti-tissue transglutaminase antibody inhibits apoptotic cell clearance by macrophages in pregnant NOD mice. *J Reprod Immunol* 2014;**103**:59–66.
18. Van Herck JL, Schrijvers DM, De Meyer GR, Martinet W, Van Hove CE, Bult H, Vrints CJ, Herman AG. Transglutaminase 2 deficiency decreases plaque fibrosis and increases plaque inflammation in apolipoprotein-E-deficient mice. *J Vasc Res* 2010;**47**:231–240.
19. Yurdagul A Jr, Doran AC, Cai B, Fredman G, Tabas IA. Mechanisms and consequences of defective efferocytosis in atherosclerosis. *Front Cardiovasc Med* 2017;**4**:86.
20. Kim HY, Kim S, Chung DH. FcgammaRIII engagement provides activating signals to NKT cells in antibody-induced joint inflammation. *J Clin Invest* 2006;**116**:2484–2492.
21. Li Y, To K, Kanellakis P, Hosseini H, Deswaerte V, Tipping P, Smyth MJ, Toh BH, Bobik A, Kyaw T. CD4+ natural killer T cells potently augment aortic root atherosclerosis by perforin- and granzyme B-dependent cytotoxicity. *Circ Res* 2015;**116**:245–254.
22. Clavel C, Nogueira L, Laurent L, Iobagiu C, Vincent C, Sebbag M, Serre G. Induction of macrophage secretion of tumor necrosis factor alpha through Fcgamma receptor IIa engagement by rheumatoid arthritis-specific autoantibodies to citrullinated proteins complexed with fibrinogen. *Arthritis Rheum* 2008;**58**:678–688.
23. Vogelpoel LT, Baeten DL, de Jong EC, den Dunnen J. Control of cytokine production by human fc gamma receptors: implications for pathogen defense and autoimmunity. *Front Immunol* 2015;**6**:79.
24. Fernández N, Renedo M, García-Rodríguez C, Sánchez Crespo M. Activation of monocytic cells through Fc gamma receptors induces the expression of macrophage-inflammatory protein (MIP)-1 alpha, MIP-1 beta, and RANTES. *J Immunol* 2002;**169**:3321–3328.
25. Taub DD, Conlon K, Lloyd AR, Oppenheim JJ, Kelvin DJ. Preferential migration of activated CD4+ and CD8+ T cells in response to MIP-1 alpha and MIP-1 beta. *Science* 1993;**260**:355–358.
26. Wojciechowski W, Harris DP, Sprague F, Mousseau B, Makris M, Kusser K, Honjo T, Mohrs K, Mohrs M, Randall T, Lund FE. Cytokine-producing effector B cells regulate type 2 immunity to *H. polygyrus*. *Immunity* 2009;**30**:421–433.
27. Dubois B, Massacrier C, Vanbervliet B, Fayette J, Briere F, Banchereau J, Caux C. Critical role of IL-12 in dendritic cell-induced differentiation of naive B lymphocytes. *J Immunol* 1998;**161**:2223–2231.
28. Enoksson SL, Grasset EK, Hagglof T, Mattsson N, Kaiser Y, Gabrielsson S, McGaha TL, Scheynius A, Karlsson MC. The inflammatory cytokine IL-18 induces self-reactive innate antibody responses regulated by natural killer T cells. *Proc Natl Acad Sci U S A* 2011;**108**:E1399–E1407.

29. Reif K, Ekland EH, Ohl L, Nakano H, Lipp M, Forster R, Cyster JG. Balanced responsiveness to chemoattractants from adjacent zones determines B-cell position. *Nature* 2002;**416**:94–99.
30. De Silva NS, Klein U. Dynamics of B cells in germinal centres. *Nat Rev Immunol* 2015;**15**:137–148.
31. Minnich M, Tagoh H, Bonelt P, Axelsson E, Fischer M, Cebolla B, Tarakhovsky A, Nutt SL, Jaritz M, Busslinger M. Multifunctional role of the transcription factor Blimp-1 in coordinating plasma cell differentiation. *Nat Immunol* 2016;**17**: 331–343.
32. Tay C, Liu YH, Kanellakis P, Kallies A, Li Y, Cao A, Hosseini H, Tipping P, Toh BH, Bobik A, Kyaw T. Follicular B cells promote atherosclerosis via T cell-mediated differentiation into plasma cells and secreting pathogenic immunoglobulin G. *Arterioscler Thromb Vasc Biol* 2018;**38**:e71–e84.
33. Sorensen MV, Pedersen S, Mogelvang R, Skov-Jensen J, Flyvbjerg A. Plasma high-mobility group box 1 levels predict mortality after ST-segment elevation myocardial infarction. *JACC Cardiovasc Interv* 2011;**4**:281–286.
34. Sudakov NP, Apartsin KA, Lepekhova SA, Nikiforov SB, Katyshev AI, Lifshits GI, Vybivantseva AV, Konstantinov YM. The level of free circulating mitochondrial DNA in blood as predictor of death in case of acute coronary syndrome. *Eur J Med Res* 2017;**22**:1.
35. Karataş R, Çelik M, Yılmaz A, Keles F, Aygül N, Vatansev H, Sökmen E, Erseçgin A. The value of heat shock protein (HSP) 60 on in-hospital and shortterm prognosis in patients with acute ST segment elevation myocardial infarction. *J Surg Med* 2020;**4**:130–134.
36. Ndungu FM, Cadman ET, Coulcher J, Nduati E, Couper E, Macdonald DW, Ng D, Langhorne J. Functional memory B cells and long-lived plasma cells are generated after a single *Plasmodium chabaudi* infection in mice. *PLoS Pathog* 2009;**5**: e1000690.
37. He M, Bianchi ME, Coleman TR, Tracey KJ, Al-Abed Y. Exploring the biological functional mechanism of the HMGB1/TLR4/MD-2 complex by surface plasmon resonance. *Mol Med* 2018;**24**:21.
38. Cohen-Sfady M, Nussbaum G, Pevsner-Fischer M, Mor F, Carmi P, Zanin-Zhorov A, Lider O, Cohen IR. Heat shock protein 60 activates B cells via the TLR4-MyD88 pathway. *J Immunol* 2005;**175**:3594–3602.
39. Bao W, Xia H, Liang Y, Ye Y, Lu Y, Xu X, Duan A, He J, Chen Z, Wu Y, Wang X, Zheng C, Liu Z, Shi S. Toll-like receptor 9 can be activated by endogenous mitochondrial DNA to induce podocyte apoptosis. *Sci Rep* 2016;**6**:22579.
40. Deguine J, Barton GM. MyD88: a central player in innate immune signaling. *F1000Prime Rep* 2014;**6**:97.
41. Khodadadi L, Cheng Q, Radbruch A, Hiepe F. The maintenance of memory plasma cells. *Front Immunol* 2019;**10**:721.
42. Kyaw T, Tay C, Khan A, Dumouchel V, Cao A, To K, Kehry M, Dunn R, Agrotis A, Tipping P, Bobik A, Toh BH. Conventional B2 B cell depletion ameliorates whereas its adoptive transfer aggravates atherosclerosis. *J Immunol* 2010;**185**: 4410–4419.
43. Iseme RA, McEvoy M, Kelly B, Agnew L, Walker FR, Handley T, Oldmeadow C, Attia J, Boyle M. A role for autoantibodies in atherogenesis. *Cardiovasc Res* 2017; **113**:1102–1112.
44. Sherer Y, Shoenfeld Y. Mechanisms of disease: atherosclerosis in autoimmune diseases. *Nat Clin Pract Rheumatol* 2006;**2**:99–106.
45. Lightman SM, Utley A, Lee KP. Survival of long-lived plasma cells (LLPC): piecing together the puzzle. *Front Immunol* 2019;**10**:965.
46. Yin C, Mohanta SK, Srikakulapu P, Weber C, Habenicht AJ. Artery tertiary lymphoid organs: powerhouses of atherosclerosis immunity. *Front Immunol* 2016; **7**:387.
47. Tay C, Liu YH, Hosseini H, Kanellakis P, Cao A, Peter K, Tipping P, Bobik A, Toh BH, Kyaw T. B-cell-specific depletion of tumour necrosis factor alpha inhibits atherosclerosis development and plaque vulnerability to rupture by reducing cell death and inflammation. *Cardiovasc Res* 2016;**111**:385–397.
48. Hermans IF, Silk JD, Gileadi U, Salio M, Mathew B, Ritter G, Schmidt R, Harris AL, Old L, Cerundolo V. NKT cells enhance CD4+ and CD8+ T cell responses to soluble antigen in vivo through direct interaction with dendritic cells. *J Immunol* 2003;**171**:5140–5147.
49. Kyriakakis E, Cavallari M, Andert J, Philippova M, Koella C, Bochkov V, Erne P, Wilson SB, Mori L, Biedermann BC, Resink TJ, De Libero G. Invariant natural killer T cells: linking inflammation and neovascularization in human atherosclerosis. *Eur J Immunol* 2010;**40**:3268–3279.
50. van Dijk RA, Duinisveld AJ, Schaapherder AF, Mulder-Stapel A, Hamming JF, Kuiper J, de Boer OJ, van der Wal AC, Kolodgie FD, Virmani R, Lindeman JH. A change in inflammatory footprint precedes plaque instability: a systematic evaluation of cellular aspects of the adaptive immune response in human atherosclerosis. *J Am Heart Assoc* 2015;**4**:e001403.
51. Biessen EA, Sluimer JC. Staging lymphocyte presence in human atherosclerosis: a tale told by numbers. *J Am Heart Assoc* 2015;**4**:
52. Arkatkar T, Du SW, Jacobs HM, Dam EM, Hou B, Buckner JH, Rawlings DJ, Jackson SW. B cell-derived IL-6 initiates spontaneous germinal center formation during systemic autoimmunity. *J Exp Med* 2017;**214**:3207–3217.
53. Browne EP, Littman DR. Myd88 is required for an antibody response to retroviral infection. *PLoS Pathog* 2009;**5**:e1000298.
54. Tian M, Hua Z, Hong S, Zhang Z, Liu C, Lin L, Chen J, Zhang W, Zhou X, Zhang F, DeFranco AL, Hou B. B cell-intrinsic MyD88 signaling promotes initial cell proliferation and differentiation to enhance the germinal center response to a virus-like particle. *J Immunol* 2018;**200**:937–948.
55. Neves P, Lampropoulou V, Calderon-Gomez E, Roch T, Stervbo U, Shen P, Kuhl AA, Loddenkemper C, Haury M, Nedospasov SA, Kaufmann SH, Steinhoff U, Calado DP, Fillatreau S. Signaling via the MyD88 adaptor protein in B cells suppresses protective immunity during *Salmonella typhimurium* infection. *Immunity* 2010;**33**:777–790.
56. Yang D, Han Z, Oppenheim JJ. Alarmins and immunity. *Immunol Rev* 2017;**280**: 41–56.
57. Almanzar G, Ollinger R, Leuenberger J, Onestingel E, Rantner B, Zehm S, Cardini B, van der Zee R, Grundtman C, Wick G. Autoreactive HSP60 epitope-specific T-cells in early human atherosclerotic lesions. *J Autoimmun* 2012;**39**:441–450.
58. Zhao Y, Zhang C, Wei X, Li P, Cui Y, Qin Y, Wei X, Jin M, Kohama K, Gao Y. Heat shock protein 60 stimulates the migration of vascular smooth muscle cells via Toll-like receptor 4 and ERK MAPK activation. *Sci Rep* 2015;**5**:15352.
59. Kanellakis P, Agrotis A, Kyaw TS, Koulis C, Ahrens I, Mori S, Takahashi HK, Liu K, Peter K, Nishibori M, Bobik A. High-mobility group box protein 1 neutralization reduces development of diet-induced atherosclerosis in apolipoprotein e-deficient mice. *Arterioscler Thromb Vasc Biol* 2011;**31**:313–319.
60. Kalinina N, Agrotis A, Antropova Y, DiVitto G, Kanellakis P, Kostolias G, Ilyinskaya O, Tararak E, Bobik A. Increased expression of the DNA-binding cytokine HMGB1 in human atherosclerotic lesions: role of activated macrophages and cytokines. *Arterioscler Thromb Vasc Biol* 2004;**24**:2320–2325.
61. Ding Z, Liu S, Wang X, Khaidakov M, Dai Y, Mehta JL. Oxidant stress in mitochondrial DNA damage, autophagy and inflammation in atherosclerosis. *Sci Rep* 2013;**3**:1077.
62. Yu E, Calvert PA, Mercer JR, Harrison J, Baker L, Figg NL, Kumar S, Wang JC, Hurst LA, Obaid DR, Logan A, West NE, Clarke MC, Vidal-Puig A, Murphy MP, Bennett MR. Mitochondrial DNA damage can promote atherosclerosis independently of reactive oxygen species through effects on smooth muscle cells and monocytes and correlates with higher-risk plaques in humans. *Circulation* 2013; **128**:702–712.
63. Pinto AR, Ilinykh A, Ivey MJ, Kuwabara JT, D'Antoni ML, Debuque R, Chandran A, Wang L, Arora K, Rosenthal NA, Tallquist MD. Revisiting cardiac cellular composition. *Circ Res* 2016;**118**:400–409.
64. Ebringer A, Rosenbaum M, Pincus N, Doyle AE. Changes in serum immunoglobulin after myocardial infarction. *Am J Med* 1971;**50**:297–301.
65. Dhamoon MS, Sciacca RR, Rundek T, Sacco RL, Elkind MS. Recurrent stroke and cardiac risks after first ischemic stroke: the Northern Manhattan Study. *Neurology* 2006;**66**:641–646.
66. Boulanger M, Bejot Y, Rothwell PM, Touze E. Long-term risk of myocardial infarction compared to recurrent stroke after transient ischemic attack and ischemic stroke: systematic review and meta-analysis. *J Am Heart Assoc* 2018;**7**: e007267.
67. Mangano DT, Browner WS, Hollenberg M, London MJ, Tubau JF, Tateo IM. Association of perioperative myocardial ischemia with cardiac morbidity and mortality in men undergoing noncardiac surgery. The Study of Perioperative Ischemia Research Group. *N Engl J Med* 1990;**323**:1781–1788.
68. Ng JLW, Chan MTV, Gelb AW, Warner DS. Perioperative stroke in noncardiac, nonneurosurgical surgery. *Anesthesiology* 2011;**115**:879–890.
69. Santos-Zas I, Lemarie J, Tedgui A, Ait-Oufella H. Adaptive immune responses contribute to post-ischemic cardiac remodeling. *Front Cardiovasc Med* 2018;**5**:198.
70. Ankeny DP, Guan Z, Popovich PG. B cells produce pathogenic antibodies and impair recovery after spinal cord injury in mice. *J Clin Invest* 2009;**119**:2990–2999.
71. Leaf IA, Nakagawa S, Johnson BG, Cha JJ, Mittelsteadt K, Guckian KM, Gomez IG, Altemeier WA, Duffield JS. Pericyte MyD88 and IRAK4 control inflammatory and fibrotic responses to tissue injury. *J Clin Invest* 2016;**127**:321–334.
72. Chen YC, Bui AV, Diesch J, Manasseh R, Hausding C, Rivera J, Haviv I, Agrotis A, Htun NM, Jowett J, Hagemeyer CE, Hannan RD, Bobik A, Peter K. A novel mouse model of atherosclerotic plaque instability for drug testing and mechanistic/therapeutic discoveries using gene and microRNA expression profiling. *Circ Res* 2013;**113**:252–265.
73. Van der Donckt C, Van Herck JL, Schrijvers DM, Vanhoutte G, Verhoye M, Blockx I, Van Der Linden A, Bauters D, Lijnen HR, Sluimer JC, Roth L, Van Hove CE, Fransen P, Knaapen MW, Hervent AS, De Keulenaer GW, Bult H, Martinet W, Herman AG, De Meyer GR. Elastin fragmentation in atherosclerotic mice leads to intraplaque neovascularization, plaque rupture, myocardial infarction, stroke, and sudden death. *Eur Heart J* 2015;**36**:1049–1058.